The Record and the Account

What a Track Record Can Show About Knowing the Chances

2026-10-21

Can a forecaster's track record show that she knows the chances? Results in economics show that if a test of forecasts is sure to pass anyone who announces the true chances, whatever they are, then someone who knows nothing, but knows the test, can pass it too, on every possible run of outcomes. I argue that the right lesson is not that knowledge of chance is untestable but that every test of it is relative to an account, fixed by whoever reads the record, of what the chances could be; and that a record can tell knowers apart only insofar as the hypotheses the account admits can be told apart by records. An account on which every mixture of chances is itself a chance certifies no one. The point holds for sincere forecasters as well as strategic ones, because knowledge of chance can be learned from the very record that is meant to test it: what a record measures is a head start over some learner, and which learner sets the standard is part of the account. Whether the world itself supplies the account depends on the metaphysics of chance. On propensity views it does not. On best-system views it does only if the standards of simplicity exclude mixtures, and then either the standards are ours, so that the relativity is part of what chance is, or they are not, and knowledge of best-system chance is testable where knowledge of primitive chance is not. The upshot for the novice is that a record can show her whom to defer to, relative to her own account, but cannot show her who knows; and the upshot for institutions is that a claim to have validated forecasts by their record is incomplete until it states the account.

Two guides

Socrates tells Meno that a man with a true opinion about the road to Larissa will guide others there as well as a man who has been down the road and knows it; for getting people to Larissa, true opinion is no worse than knowledge (*Meno* 97a–c).[1] What makes knowledge worth more is that it is tied down by an account of the reason why (98a), and later in the

[1] References to Plato are by Stephanus page; see Plato (1997).

dialogue the good statesmen of Athens are said to have done well by true opinion, "like soothsayers and prophets" who say many true things without knowing what they say (99b–d). In the *Charmides* the same thought yields a problem for anyone choosing a guide. To tell a doctor from a man who only pretends to be one, you must examine what he says and does about health and disease; a person who knows only *that* someone has knowledge, and not medicine, cannot make the examination; a fellow practitioner can (170d–171c).

The modern form of Socrates' problem is the novice/expert problem: how can someone without expertise in a domain rationally decide whom to trust in it? Goldman's influential answer lists five kinds of evidence available to the novice, the fifth of which is the candidate's track record (Goldman 2001). Track records seem to evade the Socratic difficulty. I cannot examine the astronomer's reasoning, but I can check whether the eclipse came when she said it would. Recent critics doubt that novices can use this evidence well, since records are disputed and judging their relevance itself takes expertise; Levy concludes that "we actually deploy the criteria to extend our trust, rather than in a way that could ground it" (Levy 2025, 2089). But the doubt concerns access to records. That a good enough record would settle who knows is rarely questioned.

I question it for an important kind of expertise: knowing the chances. Weather forecasters, clinicians giving prognoses, election modellers and actuaries are experts, when they are, because the probabilities they assign match the chances. For this kind of expertise there is a body of theorems in economics, so far ignored by philosophers, which says that if a test of forecasts is sure to pass anyone who announces the true chances, whatever they are, then someone who knows nothing but knows the test can pass it too (Sandroni 2003; Olszewski and Sandroni 2008; Shmaya 2008). The economists have drawn morals of their own: that falsifiability cannot screen strategic experts (Olszewski and Sandroni 2011), and that "the only coherent notion of 'true' probabilities is relative" (Al-Najjar and Weinstein 2008, 555). My aim is to say what the relativity is relative to, why it is not confined to strategic forecasters, and what it means for the metaphysics of chance and for anyone who has to choose whom to believe. The thesis is this:

> A record of forecasts – the forecasts made, and the outcomes that followed – can show that someone knows the chances only relative to an account, fixed by whoever reads the record, of what the chances could be; and it can tell knowers apart only insofar as the hypotheses that account admits can be told apart by records. Accounts on which mixtures of chances are chances certify no one.

"Account" here has the sense of Meno's geometers, who reason from a hypothesis about what the figure could be (86e–87b), not the sense of 98a, where the account is the knower's own grasp of the reason why. By looking only at records I deliberately set aside the place where Plato locates the difference between knowledge and true opinion, the knower's account; the question is what can be said by someone who cannot examine it.

Section states the theorems and their limits. Section says what an account is, why its work is done by a single formal property, and why the conclusion holds for sincere forecasters.

Section asks whether the metaphysics of chance supplies the account. Section turns to comparison between forecasters, Section to forecasters who know the test, and Section to what is left for the novice and for institutions.

What the theorems say

A forecaster watches outcomes one at a time, rain or no rain each day, and before each one announces a probability. A *test* is any rule that takes forecasts and outcomes and returns a verdict. Two standards will be used throughout, and kept apart. A test has *type-I control* if a forecaster whose forecasts are the true conditional chances passes with probability at least $1-\varepsilon$ under those chances, for every possible chance process. A test can be *ignorantly passed* if some randomised forecasting strategy, using no knowledge of the process, passes with probability at least $1-\varepsilon$ over its own randomisation on every possible sequence of outcomes.

Calibration is the familiar test: of the days a forecaster calls 30%, about 30% should be rainy. It has type-I control, by the law of large numbers. It is also cheap. A forecaster who says 30% every day where it rains on 30% of days is calibrated and useless (Seidenfeld 1985); and Foster and Vohra (1998) showed that a forecaster who knows nothing about the weather can, by randomising his forecasts in response to his past calibration errors, be calibrated on every possible sequence, even one chosen by an adversary who knows his method but not his coin flips.

The difficulty is not peculiar to calibration. Sandroni (2003) showed that every test on records of a fixed finite length that has type-I control can be ignorantly passed. Olszewski and Sandroni (2008) extended this to tests on unbounded records that reject at a finite time, provided that whether a forecaster is rejected at a time depends only on his forecasts about data already observed; Shmaya (2008) extended it to every Borel test that looks only at the forecasts actually made, even one that reserves its verdict until infinity. These results concern tests that respect Dawid's *prequential* principle, that forecasts be assessed only through the forecasts made and the outcomes that occurred (Dawid 1984). Finally, Olszewski and Sandroni (2009c) showed that once there is a bound on how much data the tester will wait for, every test with type-I control over a convex and compact class of processes can be ignorantly passed, including tests that require the forecaster to commit in advance to his whole theory.

Three features of the results matter below.

The engine. The proofs turn on a minimax theorem. Let Nature choose a process and the forecaster choose a strategy, the forecaster being paid when he passes. Against any process the forecaster has a good reply: announce it. Against any *lottery* over processes he also has one, because a lottery over processes is observationally a single process, the mixture, which he can announce. Al-Najjar and Weinstein call this the "reducibility of Nature's compound lotteries" and identify it as the key (2008, sec. 6). The minimax theorem then delivers one randomised strategy that does well against every process at once, and so on every sequence.

Type-I control "for every process" is itself a commitment about what the chances could be: the maximal one, closed under mixture.

Who the charlatan is. He needs to know the test; no single strategy passes every test of the relevant kind at once (Olszewski and Sandroni 2008). He need know nothing about the weather, and with bounded data he need not adapt as the outcomes come in: a single randomisation, at the start, over complete theories suffices. His forecasts are not secretly good; they differ from the chances, and only the verdicts coincide. A *sincere* forecaster, who forecasts by a fixed method of his own, is not the theorems' target. Section argues that the thesis covers him anyway.

What escapes. Some tests cannot be ignorantly passed. They violate the prequential principle, using the forecaster's predictions about histories that have not occurred, and they need unbounded data (Dekel and Feinberg 2006; Olszewski and Sandroni 2009a). Even then they show only that the charlatan *can* fail: for any probability measure fixed in advance, some strategy passes on all but a set of sequences of small measure (Olszewski and Sandroni 2009c, Prop. 3). No record of forecasts actually made, finite or infinite, is a test of this kind, and I set them aside.

Accounts

What an account is

Suppose the tester counts only i.i.d. processes as possible chance processes and requires the forecaster to announce, in advance, the bias he takes the coin to have. A test that fails him if the observed frequency strays far from his announced bias has type-I control for every bias, and no lottery over biases passes it on every sequence, because no lottery puts high probability near every frequency at once. The charlatan's natural reply to a mixture of biases, which is to announce the mixture, is not open to him: the mixture is not an i.i.d. process, so by the tester's lights it is not a possible chance process. Reducibility fails, and so does the impossibility.

What has the tester brought? Not merely a credence about outcomes. If she is unsure which bias obtains, her credence about outcomes is itself a mixture of i.i.d. processes, and a decision-maker who evaluates options by their worst expected value over a set of probability functions cannot distinguish that set from its convex hull. The restriction concerns how her uncertainty is carved into hypotheses about the chances. Call any such restriction an *account* of what the chances could be.

The work an account does is captured by a single property. Two hypotheses about a process are *mutually singular* if there is an event, determined by the infinite sequence of outcomes, which the first gives probability one and the second probability zero; they are then told apart, with certainty, by an unbounded record. Kavaler and Smorodinsky prove that a test comparing forecasters that uses only the forecasts made and is guaranteed to pick out whichever

of them has the true chances exists for a class of hypotheses if and only if the hypotheses in the class are pairwise mutually singular; when one hypothesis is absolutely continuous with respect to another, any test that never misidentifies the knower must sometimes be inconclusive (2019, Thm. 5, Prop. 1). Distinct i.i.d. biases are mutually singular, since the limiting frequency tells them apart. A mixture of two hypotheses is never singular with respect to its components: whatever either component makes certain, the mixture gives positive probability. So an account closed under mixture contains hypotheses no record can separate, which is the formal core of the impossibility. On finite records singularity becomes a matter of rates, how quickly the likelihood ratio between hypotheses grows, but the shape of the result is the same.

So the thesis can be put precisely. What a record certifies is fixed by the account its reader brings, and the account certifies knowers only to the extent that the hypotheses it admits are mutually singular. In the vocabulary of formal learning theory, an account fixes a space of possibilities and a topology on it, and what statistical data can verify is a neighbourhood of the truth, never the point hypothesis that *these* are the chances, unless that hypothesis is isolated among those the account admits (Genin and Kelly 2017).

Sincere forecasters: the record as textbook

The theorems are about forecasters who know the test. A novice choosing between two sincere doctors faces no such forecaster. Is the thesis, for her, anything more than the old point that evidence supports a hypothesis only relative to alternatives?

It is, because of something special about knowledge of chance: it can be learned from the very record that is meant to test it. A forecaster who does not know the bias of a coin, but forecasts by Laplace's rule of succession, comes to forecast nearly as the knower does, since the rule converges to the true bias whatever it is. More generally, whenever the true process is absolutely continuous with respect to a learner's prior, the learner's forecasts merge with the true chances as the record grows (Blackwell and Dubins 1962). A record of forecasts is both the test of the forecaster's knowledge and the textbook from which anyone could acquire it. So what a record of sincere forecasts measures is not knowledge as against ignorance but a *head start* over learners, and how large a head start counts depends on which learner sets the standard.

The dependence is sharp. If the learner's prior gives the true process positive weight w – if the learner already entertains it as a live possibility – then on every sequence the knower's likelihood ratio against the learner is at most $1/w$, and the record never tells them apart by more than that. If the learner's prior is diffuse, as Laplace's is over biases, the truth is singular with respect to it, and the knower's likelihood ratio grows, but slowly: on average, roughly as the square root of the number of observations for each unknown parameter (Clarke and Barron 1990). A sincere forecaster who happens to put most of his prior weight on the truth is Meno's guide with a true opinion; no record distinguishes him from the knower, and none

should. The record certifies knowledge of chance only as a margin over a baseline learner, and choosing the baseline is choosing an account.

The evaluator's own forecasts

The natural baseline for an evaluator is herself. Stewart (2011) gives a test that uses nothing but her own forecasts: pass a forecaster if, in the long run, his forecasts are more likely than hers on the data and differ from hers non-negligibly often. If she is a Bayesian, a forecaster who knows the true process passes with probability $1 - \varepsilon$ by her lights, where ε is her prior probability that her own forecasts would have converged to the truth quickly without help; and any forecaster who knows no more than she does passes with probability zero, again by her lights.

The test is relative to her forecasts, but what a pass shows about *knowing the chances* is relative to how she carves those forecasts into chance hypotheses. Suppose she forecasts one half every day. She may do so because she is certain the process is a fair coin; then $\varepsilon = 1$, no one can add to what she knows, and the test fails every true expert. Or she may do so because she is uniformly uncertain which deterministic sequence will occur, a state with exactly the same forecasts; then $\varepsilon = 0$, and a pass certifies a prophet. The verdicts are the same in both cases; what differs is whom she counts as a knower. A de Finettian will say that the carving is idle and that the test certifies only that the forecaster had information she lacked (Finetti 1937). That is the thesis in subjectivist dress: drop the chances and certification is relative to the reader's predictive probabilities. Those who keep the chances must say which carving they mean, and that is an account.

Two costs should be noted. The guarantee holds by her lights: she is certain that charlatans fail, and in a world her prior treats as negligible someone with no special knowledge but a better prior will pass. And when her prior would eventually have led her to the truth, what is certified is speed; Pomatto, Al-Najjar, and Sandroni (2014) tie testability to the merging of opinions in exactly this way.

Does the world supply the account?

If the metaphysics of chance fixes which probability functions are possible chance functions, perhaps it fixes the account, and records can certify knowledge of the chances simpliciter. To ask this fairly the two main views must be held to the same standard of type-I control. For views on which there is a fact about the chances independent of the whole history, the standard is the one in Section . For the best-system view the fair analogue uses the New Principle: a forecaster who announces a system S should pass with probability at least $1 - \varepsilon$ according to S, conditional on S's being the best system (Lewis 1994; Hall 1994).

Propensities. On propensity and other primitivist views, chances are single-case physical tendencies, and mixtures of chance processes can be chance processes. Pólya's urn is the standard case: draw a ball, return it with another of the same colour, repeat. The urn's chance of red at each draw is fixed by its current composition, which is a function of the draws so far; there is no hidden parameter. Yet the process is an exchangeable mixture of i.i.d. processes, with a Beta mixing distribution, by de Finetti's theorem (Hill, Lane, and Sudderth 1987). Nothing about the example is special: a mechanism whose state records the outcomes so far, and which at each step produces an outcome with a chance computed from that state, can realise any process whatever as its chance process. So on these views the class of possible chance processes is closed under mixture, reducibility holds, and no record certifies anyone. Indeed the urn and a coin whose unknown bias was fixed once and for all give the same probabilities to every sequence of outcomes, so not even the whole history distinguishes them. This is the analogue, for chance, of the epistemological argument of Earman and Roberts (2005a, 2005b), who argue that whatever makes a difference to what the laws are must supervene on all the possible evidence (they leave chance aside): a difference in chance that makes no difference to the probabilities of outcomes is inaccessible even to total evidence.

Best systems. On the best-system analysis, the chances are given by the system that best balances simplicity, strength and fit, where fit is the probability the system gives the actual history (Lewis 1994; Loewer 2004). A mixture of two systems fits no better than its better component, since its fit is a weighted average of theirs. Whether a mixture can ever be best therefore turns on simplicity. If the standards of simplicity penalise mixing systems that the history could tell apart, then, for instance, the half-and-half mixture of a 30% coin and a 70% coin is never best, the candidate systems that could be best for histories like ours form a class that is not closed under such mixtures, and under the New Principle standard knowers of the best system can in principle be told apart from those who merely announce mixtures. But not every way of measuring simplicity does this. The chance functions of statistical mechanics are mixtures over deterministic histories and are best because they are simple (Loewer 2004); and on description-length measures of simplicity, which have been recommended for the best-system view (Schwarz 2016), the parameter-free Laplace rule, which is Pólya's urn again, competes with every coin of fixed bias (Grünwald 2007).

So the best-system view supplies a certifying account only if its standards of simplicity exclude the relevant mixtures, and this yields a dilemma. If the standards are ours – if, as Lewis allowed it might seem, they are partly a matter of our psychology, so that the best system is best under any standards only if nature is kind (Lewis 1994, 479), or if, as on Hoefer's pragmatic version, the best system is the one best for agents like us (Hoefer 2007) – then the account that makes chance-knowledge testable is ours after all, and the relativity of the thesis is not a limit on knowing the chances but part of what the chances are. If the standards are objective, then knowledge of best-system chance can be certified, in principle, by a record and an appropriately stated test, and knowledge of primitive chance cannot. That is a new epistemic consideration in the debate over the metaphysics of chance. Lewis's complaint against primitive chance was that he did not "begin to see" how knowledge of it could constrain rational

credence (Lewis 1994, 484); the complaint here is that nothing in any record could show that anyone knows it.

Computation. A third candidate is computability. If the charlatan must compute his forecasts efficiently there are tests he cannot pass (Fortnow and Vohra 2009; Hu and Shmaya 2013), and if the chances are computable, any two computable forecasting systems that are calibrated with respect to every computable way of selecting days must agree in the limit (Dawid 1985). That is identification from the record alone, relative to a computational class. But the agreement is asymptotic: on any finite record two such systems can disagree arbitrarily, and Belot (2013) shows that topologically typical sequences have no computable calibrated forecaster at all.

Comparison

The evaluator is one baseline; another forecaster is a second. Here the formal results bear out Socrates and also correct a natural reading of him.

Suppose B's forecasts are the true chances, and A is any rival whose forecast each day is fixed before the outcome. Under the true chances the ratio of the probabilities A and B gave to what happened is a nonnegative martingale with mean one, so the probability that A ever outscores B by a factor c is at most $1/c$, whatever A does (Ville 1939). A known craftsman, then, cannot be beaten except by luck. What he can do is discredit rivals, since a rival whose forecasts stray substantially from the chances will typically be outscored; and a rival who is not beaten is shown to forecast close to him (Al-Najjar and Weinstein 2008, Thm. 1). This is how the fellow practitioner of the *Charmides* picks out the real doctor: by finding that what the man says agrees with what he himself knows. He certifies agreement and discredits by beating; he does not crown anyone as knowing more than he does.

Crowning a winner needs a different premise: that one of two forecasters knows the chances, though the evaluator does not know which. If A then outscores B by a factor c, A is the knower unless an event of probability at most $1/c$ has occurred. This is the case for which Al-Najjar and Weinstein designed their comparative test, and Feinberg and Stewart (2008) give an infinite-horizon analogue. What is certified is relative to the pair: given that one of us knows, it is A.

Comparison has two limits that records cannot remove.

Provenance. Comparative certification assumes that each forecaster's forecasts express what he knows. A forecaster who sees a rival's current forecast can beat the rival's score by the rival's calibration error, without any knowledge of the domain, by recalibrating the rival's forecasts against their past performance (Foster and Hart 2023). He cannot do better than the rival's refinement, the part of the score that reflects sorting rainy days from dry, so he improves on the rival without adding knowledge of the weather; but a comparison will credit him with

knowledge that is the rival's. Whether one forecast draws on another is a fact about provenance that the record does not contain. Rules can forbid looking at rivals' current forecasts; they cannot stop two forecasters from drawing on the same model.

The guild. Can comparison at least tell whether anyone in a group knows the chances? On records of finite length, any test meant to say "at least one of these two is informed" that passes when one of them is truthful can be passed by two uninformed forecasters (Al-Najjar and Weinstein 2008, Thm. 7); independently randomising uninformed forecasters can jointly pass any comparative test of the future-independent kind with type-I control (Olszewski and Sandroni 2009b; Olszewski 2015, Prop. 8); and the cross-calibration test of Feinberg and Stewart makes joint passing by uninformed forecasters rare only by failing some true experts. A field cannot certify itself by its members' records. Where a field contains a knower, records can sort its members; in a guild of non-knowers they sort non-knowers, and nothing in the comparison reveals which situation obtains.

Forecasters who know the test

The theorems show that knowledge-free strategies passing any given test exist. Whether a forecaster uses one is an empirical question about the forecaster. That question is newly pressing because of machine forecasters, whose forecasting is evaluated on public benchmarks with published metrics (e.g. Karger et al. 2025).

Two cases should be distinguished. A model tuned on past questions and then evaluated, as a fixed function, on questions that resolve after its training is a sincere forecaster in the sense of Section : tuning against a metric is a Goodhart problem, which held-out questions address, not the charlatan's. For such a model, the analysis of Section applies: its record certifies a margin over a baseline the evaluator chooses. But a system that can be optimised against the specific test, or that adapts to its own scores, is a candidate for the charlatan's role, and then the existence of a knowledge-free passing strategy means that an absolute pass mark – a calibration threshold, or a Brier score below some level – is no evidence of knowledge unless one has independent reason to think the system is not using such a strategy. Calibration-type tests in particular can be passed by efficient procedures (Foster and Vohra 1998; Foster and Hart 2021).

Comparison with human forecasters, the usual alternative, is exposed to provenance failure when systems can see the crowd's forecasts. Copying the crowd inherits the crowd's resolution, and recalibrating it removes the crowd's calibration error; neither is knowledge of the domain. What a comparison can certify is resolution beyond the crowd's, and since the crowd is a rival and not a known knower, it certifies it only in the sense of Section : better than the crowd, on these questions.

The novice

What is left for the novice? Not certified knowledge of the chances; but that was never what she needed. She needs to know whom to defer to, and deferring is comparative. Dorst and coauthors characterise *total trust* in a source – for every quantity, expecting it to be at least t given that the source's estimate is at least t – and show that it is equivalent to being willing to let the source make any decision on one's behalf; unlike deference modelled on Reflection, it can be extended to a source that is modest about itself (Dorst et al. 2021). The positive results above certify exactly something of that comparative shape: that a forecaster's forecasts add to one's own, relative to one's own account (Section), or that one of a pair does better than the other, given that one of them knows (Section). The negative results tell only against certifying that anyone knows the chances, which no novice needed.

Even comparative certification asks a lot: an account of what the chances could be, which is much of what a novice lacks. But she can benefit from experts without certifying any of them. Suppose she forms her forecasts by mixing her own and the candidates' forecasts, weighting each by how well it has done: a Bayesian mixture over forecasters rather than over chance hypotheses. Then on every sequence of outcomes her cumulative logarithmic score is within $\log(1/w)$ of the best forecaster in the mix, where w is the weight she started him with, whatever the forecasters do; on the use of such guarantees by novices see Herrmann (2022), and for contracts that capture the value of advice without learning the adviser's type, Olszewski and Pęski (2011). This needs no account of the chances and no assumption about provenance. It is not deference, since she totally trusts no one; it is a guarantee that she does nearly as well as the best of those she listens to. Its limits are plain. It is only as good as the best forecaster in the pool, and by Section she cannot tell whether anyone in the pool knows anything. It also needs repeated forecasts that can be checked, and most of the disputes Levy has in mind, about climate or policy, are not like that.

Levy's slogan, that the criteria extend trust rather than ground it, is vindicated here in a sharper form, and for a different reason. The difficulty is not access: the theorems grant every outcome, every forecast and unlimited data. It is that what a record certifies depends on an account it does not contain. What gets extended is not trust in persons but an account of what the chances could be.

That suggests a norm for the institutions that ask the public to trust forecasts because of their record: weather services, regulators, the owners of benchmarks. A claim to have validated a forecaster by its record is incomplete until it states the account relative to which the record was read. And the account must be fixed before the record accrues. Records can discredit an account, as a forecaster can be discredited, when they fall in the far tail of what it predicts (Box 1980); but the guarantees of the tests above hold only for an account chosen in advance, and an account revised in the light of the record has used the same data twice. A novice who cannot evaluate the account can still ask whether it was stated, whether it was fixed in advance, and whether those who bear the costs of error can contest it; these are the kinds of indicators of trustworthiness that laypeople can assess (Anderson 2011).

Objections

The type-I requirement is too strong. Requiring a test to protect experts whatever the chances may seem excessive; Stewart notes that a tester facing a decision problem would impose it only if extremely averse to ambiguity (2011). I agree. The thesis is that relaxing it means supplying an account; the theorems show what happens when none is supplied.

Records can discredit. They can. By the inequality used in Section , being outscored by a factor c discredits a claim to know the chances at level $1/c$, and Shafer (2021) builds a theory of testing forecasters on this. But discrediting is not certifying, and the novice's question is whom to trust. Popper's asymmetry between refutation and verification reappears, at the level of persons (cf. Olszewski and Sandroni 2011).

Real experts cannot commit in advance. The tests that resist manipulation require forecasters to commit to predictions about histories that never occur, and the Eleatic Stranger argues that a true expert should not be bound by rules laid down in advance (*Statesman* 294a–c). The point bears on those tests, which I have set aside; it reinforces the conclusion that records of forecasts actually made do not certify.

Brier scores work in practice. They work comparatively: a forecaster with a lower Brier score than a rival forecast better on these days. That is the certification of Section , with its limits.

Conclusion

A good enough record seems as if it should settle whether someone knows the chances. It does not, and not only because someone who knows the test can pass it. Knowledge of chance is the kind of knowledge a record can teach, so a record measures a head start, and how much of a head start shows knowledge depends on an account of what the chances could be, which the reader brings. The account certifies only where its hypotheses can be told apart, and an account closed under mixture tells no one apart. The world supplies such an account only if chances are best-system chances whose standards of simplicity exclude mixtures, and then either the standards are ours or knowledge of primitive chance is untestable. Socrates' fellow practitioner brings an account in the form of his own knowledge. The rest of us should ask experts, and the institutions that certify them, to state theirs.

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